



SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

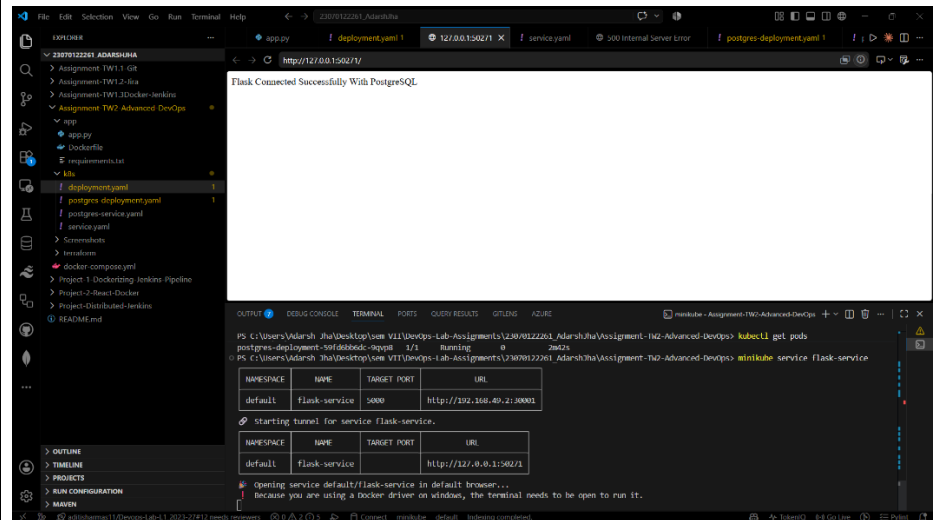
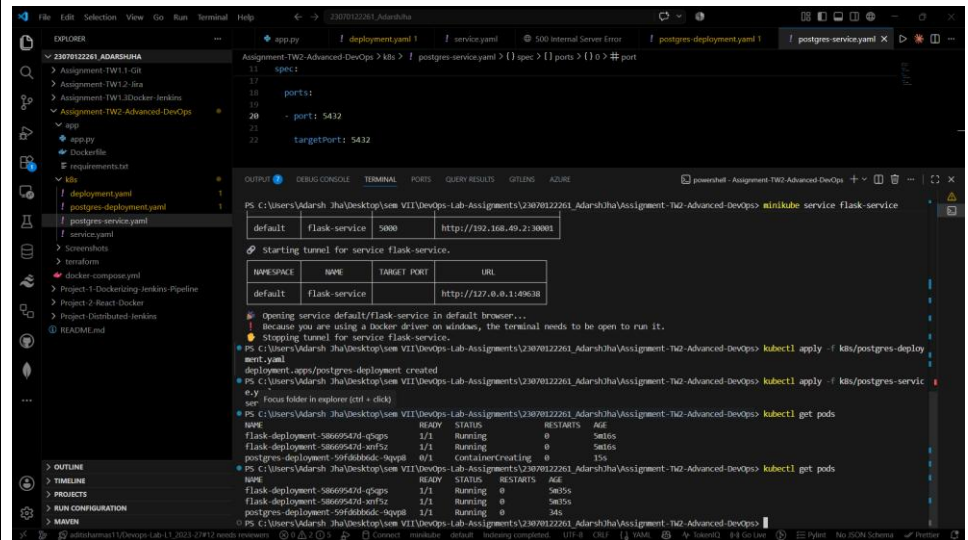
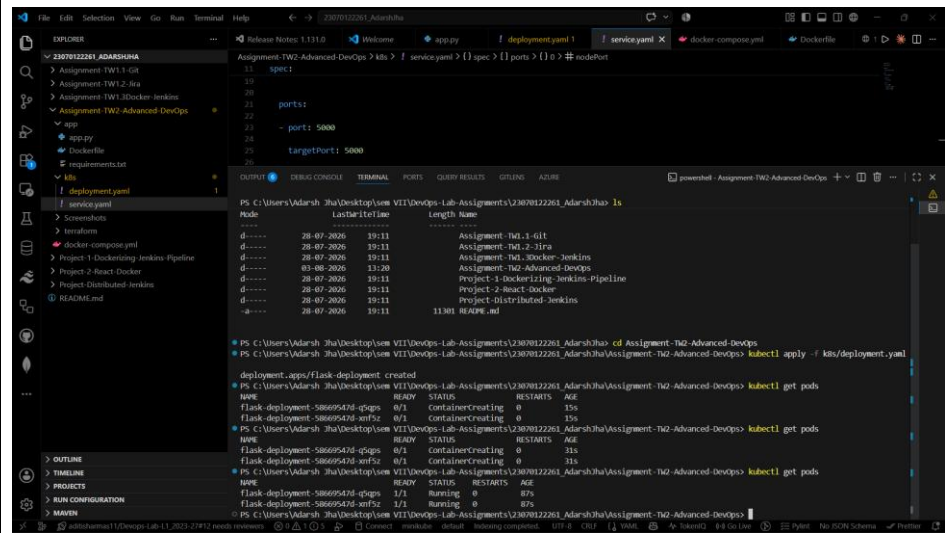
Re-accredited by NAAC with 'A' grade (3.58/4) | Awarded Category – I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Assignment No. 13

Subject:	DevOps Lab
TCODE:	TE7950
Name of Student	Adarsh Jha
PRN No.	23070122261
Branch	CSE, Batch (2023-27), L4
Academic Year & Semester	25 – 26, Sem VII
Date of Performance	23 rd July, 2026
Title of Assignment:	Kubernetes Service & Networking
Theory:	A Kubernetes Service is an abstraction that provides a stable network endpoint for accessing one or more Pods in a Kubernetes cluster. Since Pods are temporary and their IP addresses may change, a Service uses labels and selectors to route incoming traffic to the appropriate Pods. A NodePort Service exposes the application outside the cluster by allocating a port on each cluster node, allowing external access to the application. During local development, <code>kubectrl port-forward</code> can also be used to securely forward traffic from a local machine to a Kubernetes Service for testing and verification. This approach simplifies communication between applications and ensures reliable access to containerized workloads. Commands Used • Created the Kubernetes Deployment: <code>kubectrl apply -f web-deployment.yaml</code> • Verified that the Pods were running: <code>kubectrl get pods</code> • Created the Kubernetes Service: <code>kubectrl apply -f web-service.yaml</code> • Displayed the available Kubernetes Services: <code>kubectrl get service</code> • Displayed detailed information about the Service: <code>kubectrl describe service web-app-service</code> • Displayed the Service endpoints: <code>kubectrl get endpoints</code>

	• Displayed the running Pods with labels: kubectl get pods --show-labels • Displayed the application logs: kubectl logs web-app-deployment-<pod-name> • Forwarded the Kubernetes Service to the local machine: kubectl port-forward service/web-app-service 8085:80 • Deleted the Kubernetes Service: kubectl delete -f web-service.yaml • Deleted the Kubernetes Deployment: kubectl delete -f web-deployment.yaml
Implementation:	Step 1: Create the Kubernetes Service YAML Created Kubernetes Service Manifest — Created the web-service.yaml file defining a Kubernetes NodePort Service to expose the Flask web application running in the Kubernetes Pods. Step 2: Apply the Service Applied Kubernetes Service — Applied the Kubernetes Service manifest using kubectl apply, creating a NodePort Service for the deployed Flask application. Step 3: Find the Service Information Verified Kubernetes Service — Verified the Kubernetes NodePort Service and identified the assigned NodePort used to expose the Flask web application outside the Kubernetes cluster. Step 4: Access the Application Accessed Application Through Kubernetes Service — Used kubectl port-forward to forward the Kubernetes Service to the local machine and successfully accessed the Flask web application at , verifying that it was running correctly. Step 5: Clean Up Deleted Kubernetes Service — Removed the Kubernetes Service using the kubectl delete command, cleaning up the networking resources created for the application.



Subject Faculty Name: Prof. Arijit Dutta